An ideal that is not principal

Dummit–Foote, 9.1.6

Theorem 1. *The ideal (x, y) is not principal in $\mathbb{Q}[x, y]$.*

Proof. Suppose, for contradiction, that there exists $f \in \mathbb{Q}[x, y]$ such that

$$(f) = (x, y)$$

as ideals of $\mathbb{Q}[x, y]$. Here the two variables are the generators corresponding to the two indices in $\operatorname{Fin} 2$; we write them as x and y.

Let t be the indeterminate in the one-variable polynomial ring $\mathbb{Q}[t]$. Define $\mathbb{Q}$-algebra homomorphisms

$$\varphi_0 \colon \mathbb{Q}[x, y] \longrightarrow \mathbb{Q}[t], \qquad \varphi_1 \colon \mathbb{Q}[x, y] \longrightarrow \mathbb{Q}[t], \qquad \varepsilon \colon \mathbb{Q}[x, y] \longrightarrow \mathbb{Q}$$

by

$$\varphi_0(x) = t, \quad \varphi_0(y) = 0, \qquad \varphi_1(x) = 0, \quad \varphi_1(y) = t, \qquad \varepsilon(x) = \varepsilon(y) = 0.$$

Thus ε is evaluation at $(0, 0)$.

Since $x, y \in (x, y) = (f)$, there exist $a, b \in \mathbb{Q}[x, y]$ such that

$$x = fa, \qquad y = fb.$$

On the other hand, $f \in (f) = (x, y)$, so there exist $u, v \in \mathbb{Q}[x, y]$ such that

$$f = xu + yv.$$

Applying ε gives

$$\varepsilon(f) = \varepsilon(x)\varepsilon(u) + \varepsilon(y)\varepsilon(v) = 0. \tag{1}$$

Applying φ_0 to $x = fa$, φ_1 to $y = fb$, and φ_1 to $x = fa$, we obtain respectively

$$\varphi_0(f)\,\varphi_0(a) = t, \tag{2}$$

$$\varphi_1(f)\,\varphi_1(b) = t, \tag{3}$$

and

$$\varphi_1(f)\,\varphi_1(a) = 0. \tag{4}$$

Since $t \neq 0$ in $\mathbb{Q}[t]$, equations (2) and (3) imply

$$\varphi_0(f) \neq 0, \qquad \varphi_0(a) \neq 0, \qquad \varphi_1(f) \neq 0.$$

Hence from (4), since $\mathbb{Q}[t]$ is a domain and $\varphi_1(f) \neq 0$, we get

$$\varphi_1(a) = 0. \tag{5}$$

For every $p \in \mathbb{Q}[x, y]$, evaluating $\varphi_0(p)$ or $\varphi_1(p)$ at $t = 0$ amounts to evaluating p at $(0, 0)$. That is,

$$\varphi_0(p)(0) = \varepsilon(p), \qquad \varphi_1(p)(0) = \varepsilon(p). \tag{6}$$

Indeed, both assertions hold because the corresponding homomorphisms agree on the generators x and y.

By (1) and (6), the constant coefficient of $\varphi_0(f)$ is

$$\varphi_0(f)(0) = \varepsilon(f) = 0. \tag{7}$$

From (2), taking degrees of nonzero polynomials, we obtain

$$\deg \varphi_0(f) + \deg \varphi_0(a) = \deg t = 1. \tag{8}$$

If $\deg \varphi_0(f) = 0$, then $\varphi_0(f)$ would be equal to its constant coefficient, which is 0 by (7), contradicting $\varphi_0(f) \neq 0$. Therefore

$$\deg \varphi_0(f) > 0.$$

Combined with (8), this yields

$$\deg \varphi_0(a) = 0. \tag{9}$$

Since also $\varphi_0(a) \neq 0$, (9) implies that its constant coefficient is nonzero: otherwise a degree-0 polynomial with zero constant coefficient would be zero. Thus

$$\varphi_0(a)(0) \neq 0. \tag{10}$$

However, by (5) and (6) applied to $p = a$,

$$\varepsilon(a) = \varphi_1(a)(0) = 0.$$

Applying (6) again, but now with φ_0, gives

$$\varphi_0(a)(0) = \varepsilon(a) = 0,$$

contradicting (10).

This contradiction shows that no such f exists. Hence (x, y) is not principal in $\mathbb{Q}[x, y]$. □